Asymmetric packet balancing for subexponential SAT

A review of the recent ProveIt proposals and a sharper contraction bound

Research analysis prepared for Vladimir Reshetnikov
with ChatGPT

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Abstract

We review the seven analysis packets placed in ProveIt's `Computability/SubexponentialSAT` project on October 26, 2026. Their strongest justified general-density bound remains $\mathrm{poly}_k(n,m)2^{O_k(n\log\log n/\sqrt{\log n})}$, with a sharper clause-density dependence and several improvements to explicit coefficients. We prove a further refinement of their common tensor-contraction framework. A deterministic, exact-rational, two-stage labeling replaces collision variance by nearly uniform pair loads. A second improvement uses the asymmetric degrees created by the recursive child construction. Together with an additive precision buffer, the previously proved sharp Hermite budget, and factorial Cauchy modes, the resulting normalized bound is

$$T(s,\mathcal{L}) \le (s+\mathcal{L})^{O(1)}2^{(K_{\mathrm{asym}}+o(1))s/\sqrt{\log_2 s}}, \qquad K_{\mathrm{asym}} = \frac{\pi\sqrt{5/2}}{\ln 2}(9+\sqrt{3}) = 76.9089949516\ldots.$$

Here the root has at most s Boolean wires and maximum degree at most $\lceil s^{9/10}\rceil$, with the other input conditions of the cited contraction theorem. This improves the earlier explicit certificate $648.4410666488\ldots$ in the same size normalization; the comparison concerns upper bounds, not measured runtimes. A separate integral Cauchy basis exploiting the actual reciprocal row domain gives mode slopes $4, 16/5, 64/21, \ldots$, without increasing the number of mode families. We also give a $31.0641320630\ldots$ coefficient for suitably low-degree inputs. These results do not improve the power of the logarithm or remove the general-density $\log\log n$ factor. The report contains ordinary mathematical proofs and exact finite prototypes. It does not supply an end-to-end SAT solver or a new proof-assistant certificate, and its SAT consequences retain the stated upstream dependencies.

Source boundary. The preprint is pinned to `openai/math@7ef5c0cc5c`. The repository placement and comparison are pinned to `ProveIt@2b925d8800`. Full identifiers, immutable links, and file hashes are supplied in `source_manifest.json`.

Contents

1 Outcome, comparison, and meaning of the improvement **3**
1.1 What the constants compare 3
1.2 What remains unresolved 4

2 The reviewed commits and proposal frontier **4**

3 The inherited exact contraction interface **5**

4 Deterministic asymmetric balancing of pair loads **6**
4.1 Why the power-scale conditions are useful and close to sharp 7

5 Packet counts, crossing families, and the recursive asymmetry **8**

6 Precision and entropy with an additive buffer **8**
6.1 The factorial weighted-partition bound 9

7 The explicit asymmetric schedule and its global bound **10**
7.1 Applicability without a circular depth assumption 10
7.2 Summing the actual work 10

8 An additional improvement from the actual Cauchy row domain **11**

9 Consequences for SAT and integration with the earlier results **14**

10 Executable evidence and its limits **15**

11 Why the logarithmic order survives, and further research **16**

A A low-degree contraction coefficient **16**

B Optimization of the depth allocation **17**

1 Outcome, comparison, and meaning of the improvement

The requested target was an improvement beyond the recent ETH-related proposals, not merely a restatement of the cited preprint. The new contribution is a different constructive packing lemma and a recursion schedule that uses the two sides' different degree bounds. It composes with the strongest already proposed algebraic refinements.

The following theorem is the principal result. All logarithms in its normalization are base two; ln is the natural logarithm.

Theorem 1.1 (Refined contraction bound, relative to the stated framework)**.** *Let $s \geq 2$. Consider a bipartite Boolean tensor network with at most s wires, polynomially many vertices, maximum degree at most $\Delta = \lceil s^{9/10} \rceil$, and total graph/tensor description length $\mathcal{L}$. Its entries belong to*

$$(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell}), \qquad 1 \leq P_0, \ell \leq Cs,$$

for a fixed C. Suppose each original tensor entry is evaluable in $(s+\mathcal{L})^b 2^{As^{9/10}}$ time and $(s+\mathcal{L})^b$ space, for fixed A, b. Using the exact deformation and child-contraction identities of the cited preprint, there is a deterministic algorithm with polynomial space and bit time

$$(s+\mathcal{L})^{O(1)} 2^{(K_{\mathrm{asym}}+o(1))s/\sqrt{\log_2 s}}, \qquad K_{\mathrm{asym}} = \frac{\pi\sqrt{5/2}}{\ln 2}(9+\sqrt{3}). \tag{1}$$

The $o(1)$ term tends to zero for the fixed interface constants above. The algorithm may use either the earlier factorial Cauchy basis or the stronger annulus basis proved in Section 8.

The qualifier in the theorem is substantive. We prove the new packing, precision scheduling, and complexity statements here; we retain the preprint's exact algebraic contraction architecture. We inspected the relevant source identities and the earlier modifications, but did not rebuild the entire upstream formalization or independently formalize this theorem.

1.1 What the constants compare

The contraction size s counts Boolean wires. It is not automatically the number n of SAT variables. For a general formula, the reduction has $s = O_k(n \log\log n)$, with a width-dependent constant that must be tracked before converting any numerical wire coefficient into a numerical coefficient on $n \log\log n$.

Normalized contraction certificate	Base-two coefficient
Earlier source-schedule analysis in the contraction packet	18535.7538...
Earlier refined schedule [6]	648.4411...
Earlier schedule plus factorial modes, matched composition	374.4
New symmetric balancing and additive-buffer schedule	93.9291...
New asymmetric recursive schedule, Theorem 1.1	76.9090...
New low-degree specialization, Appendix A	31.0641...

The low-degree row has stronger input assumptions and is listed separately. The numerical comparisons use the common normalized interface with $P_0, \ell \leq s$; the new theorem additionally permits any fixed $P_0, \ell \leq Cs$. The matched composition in the third row uses the earlier refined schedule, whose leaf cost is already lower order, and multiplies its mode coefficient by $1/\sqrt{3}$. Its unrounded value is $374.378\ldots$ and is recorded by the verification script. It is a composition for comparison, not another historical packet. The final coefficient is about 11.86% of the previous 648.44 certificate. This is not an 8.43-fold wall-clock speedup: an exponent coefficient is being compared, and upper bounds do not establish ratios of actual running times.

1.2 What remains unresolved

For fixed clause width k, the strongest justified density envelope among the reviewed packets is still

$$\mathrm{poly}_k(n,m)\,2^{O_k\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}. \tag{2}$$

Our engine can be used inside that reduction and inside the scalar parity lift of the earlier refinement. It changes their contraction constants, not the asymptotic expression in (2). The proposed $n/\log n$ and related stronger exponents in the packets rest on explicitly missing summation or factorization interfaces. We do not treat those conditional targets as established algorithms, nor claim to have constructed their missing interfaces.

2 The reviewed commits and proposal frontier

The initially retrieved snapshot was `aa6933442eb936962de79ba6d321cd0391c450bd`. During the review, commit `2b925d8800cda68f5723b65dc1fcdaf3d8170359` placed the same delivered material under `Computability/SubexponentialSAT/Research`. We reviewed that placement and its index as well as the original archives. The arrival commits were:

- `a505bd768bce347c70e923586c5f72ad3085bf44`: three packets and the two primary manuscript PDFs;
- `64b3202ee84fc4d71fe69c34d2110c93b7da95cd`: three further refinement packets;
- `8afdbbea48d67b8d252a99e8926f3eca9649eae3`: the asymptotic analysis and barrier packet.

The placement records that these are external research artifacts, with no ProveIt Lean or Rocq development and no executed package validation at placement time [4]. Our checks below are new, scoped checks of our own components; they do not change that historical status.

Packet	Strongest proved refinement claimed in the packet	Remaining limitation
Factorial Cauchy modes [5]	Weights $1+b+2\nu_2(b!)$; leading abstract mode entropy decreases by $1/\sqrt{3}$.	Same logarithmic order; original packing retained.
Contraction improvements [6]	Leading precision $6MD/u$; explicit coefficient $648.441\ldots$; streamed packet coefficient extraction.	Same global logarithmic order; packet factor three retained.
Precision refinement [7]	Exact Hermite condition count and loss; real taper minimax calculation.	Stronger structured-mode summation is an unproved hypothesis.
Uniform density improvement [8]	Equation (2), uniformly in clause density.	No stronger unrestricted n-only exponent.
Density refinement [9]	$n\log(2+B)/\sqrt{\log n}$ for $m \le Bn$ in its stated range; precision/schedule savings.	Overlaps the preceding density result; no new logarithmic power.

Packet	Strongest proved refinement claimed in the packet	Remaining limitation
Scalar parity refinement [10]	Scalar integral lift; incidence-degree cap $k(p+a-1)$ rather than the source's $k(a+2(p-1))$.	Recursive compression requires an unsupplied compact factorization.
Asymptotic analysis [11]	Scoped obstructions and precise missing interfaces; no stronger unrestricted bound claimed.	Neither linear-wire compression nor degree-preserving recursion is constructed.

All seven analyze the same four preprint source files: the two pinned upstream commits in their manifests have identical relevant blobs. The separate 49-page manuscript credited to Claude (Anthropic), *A $2^{o(n)}$ time algorithm for 3-SAT*, states a bound with exponent $O(\min\{km/\sqrt{\log m}, k^2 n \log\log n/\sqrt{\log n}\})$ [12]. It is a different construction. We inspected its headline and scope; we do not claim an explicit constant comparison to that manuscript or certify its reported unpublished Lean development.

The OpenAI manuscript's sharper appendix gives the general exponent $n\log\log n/\sqrt{\log n}$ and fixed-density exponent $n/\sqrt{\log n}$ [1, 3]. The upstream formalization scope page states a weaker $(\log\log n)^2/\sqrt{\log n}$ numerator factor in its general bound [13]. Improving that weaker formal endpoint by quoting the manuscript would not answer the present research question.

3 The inherited exact contraction interface

A Boolean tensor network has one Boolean variable per edge, with contraction

$$Z = \sum_{x\in\{0,1\}^E} \prod_v T_v(x|_{\delta(v)}).$$

Parallel edges have distinct ports. The source works over an unramified extension $\mathcal{O}$ of $\mathbb{Z}_{(2)}$ whose residue field has polynomial size, and over $\mathcal{A} = \mathcal{O}[V]/(V^\ell)$ modulo powers of two. Valuation and divisibility are coordinatewise in the fixed free bases. The arithmetic cost is polynomial in all the displayed parameters.

At an active call let M be its wire count, $L = 2^a$ its packet length, and r its fixed alphabet size. Labels belong to $[r]^a$ on both sides; write $Q = r^a$. Every ordered pair of labels is padded to the same load H_{pad}, a multiple of L. The source's orientation construction then groups the padded slots into packets of length L. In coordinate i, the unordered pair of symbols is fixed within a packet, and bit i of a slot determines its orientation when the two symbols differ. Unused slots have no real ports.

The deformation uses the integer grid

$$S = \{-L, -L+1, \dots, L-1\}, \qquad \mathcal{C}(T) = \prod_{t\in S}(T-t), \qquad \alpha = 2^{a+1},$$

and a packet wire kernel

$$W_{xy}(T) = \frac{\mathcal{C}(T)}{\mathcal{C}'(0)(T-(y-x))}. \tag{3}$$

The denominator's linear factor cancels a numerator factor. The deformed contraction $F(T)$ has degree at most $M(2L-1)$, $F(0)$ is the desired contraction, and each $F(t+\alpha z)$ is integral. At $i = \nu_2(t)$, the specialization $W(t)$ connects only ports in the same coordinate-i group.

The nonlocal terms at $T = t + \alpha\zeta$ separate into linear source moments and target coefficients. At level i a column tile has step 2^{i+1} and length $J = D_i/2$, where $D_i = L/2^i$. The earlier factorial basis has certified weights

$$w_j = 1 + j + 2\nu_2(j!) = 3j + 1 - 2s_2(j), \qquad j \geq 0. \tag{4}$$

A tuple β contributes only when $\sum_Z w_Z \beta_Z < p_t$, where p_t is the working precision. For each tuple the source's exact factorization is

$$F(t + \alpha\zeta) = \sum_{\beta : \sum w_Z \beta_Z < p_t} \prod_{g=1}^{r} Z_{g,\beta} \pmod{2^{p_t}}. \tag{5}$$

The child for group g puts *all retained tensors from both sides* of the parent on one side, and new packet tensors on the other. A new packet tensor has degree at most $2L$.

This last structural fact drives the new schedule. Below the root, only one side can contain the original high-degree tensors. The other side is always made of fresh, much smaller packet tensors.

Remark 3.1 (Canonical representatives). Each tensor description retains its creation precision. Evaluating an old tensor later means evaluating it at that original precision and reusing its canonical integer coordinates. Canonical lifting is not a ring homomorphism. Neither the packing change nor the new mode basis licenses silently changing a tensor's defining lift. Our consequences use exactly the source's immutable-description convention.

4 Deterministic asymmetric balancing of pair loads

We now prove the central new combinatorial lemma independently of tensor algebra. It applies to multigraphs and uses exact rational arithmetic.

Theorem 4.1 (Two-stage pair-load balancing)**.** *Let a bipartite multigraph have M edges, maximum row degree at most R, and maximum column degree at most C, where R, C are positive integers. For integers $Q \geq 1$ and $h \geq 2$, there is a deterministic labeling of each side by Q labels such that every ordered label-pair load X_{ab} satisfies*

$$X_{ab} \leq \left(1 + \frac{2}{h-1}\right)\left(\frac{M}{Q^2} + \frac{R}{Q}\right) + 2(hC + 1)\ln Q. \tag{6}$$

The construction runs in polynomial bit time and space in the explicit graph size and the parameters Q, h, R, C.

Proof. First assign row vertices greedily to the label with smallest current sum of incident degrees. If W_a is the resulting stub load of row label a, then

$$W_a \leq M/Q + R.$$

Indeed, just before the final vertex placed in a bin, that bin has at most the current average load, and the vertex adds at most R.

For each column vertex v, let w_{av} count its edges to row label a. Multiplicities are counted, so $0 \leq w_{av} \leq C$ and $\sum_v w_{av} = W_a$. Give each column vertex an independent uniform random label temporarily. Put

$$q = 1 + \frac{1}{hC}, \qquad \Phi = \sum_{a,b=1}^{Q} q^{X_{ab}}.$$

For a fixed pair,

$$\mathbb{E}q^{X_{ab}} = \prod_v \left(1 + \frac{q^{w_{av}} - 1}{Q}\right).$$

Convexity of q^w on $[0, C]$ gives $q^w - 1 \le (w/C)(q^C - 1)$. Hence

$$\mathbb{E}\Phi \le Q^2 \exp\left(\frac{(q^C - 1)\max_a W_a}{QC}\right). \tag{7}$$

Assign the column labels in a fixed order. For each choice compute the conditional expectation of Φ exactly and choose a minimum. The average over the Q choices equals the old conditional expectation, so the selected choice does not increase it. At termination, $q^{\max X_{ab}} \le \Phi \le \mathbb{E}\Phi$. Taking logarithms for the analysis, not for the algorithm, yields

$$\max X_{ab} \le \frac{2\ln Q}{\ln q} + \frac{q^C - 1}{C \ln q}\frac{\max_a W_a}{Q}.$$

Now $1/\ln q \le hC + 1$, since $\ln(1 + x) \ge x/(1 + x)$. Also $q^C \le e^{1/h}$, and $e^{1/h} - 1 \le 1/(h - 1)$, so

$$\frac{q^C - 1}{C \ln q} \le \frac{h + 1/C}{h - 1} \le 1 + \frac{2}{h - 1}.$$

This proves (6), including $Q = 1$.

For the algorithmic claim, a conditional expectation is a sum of terms $q^{x_{ab}} \prod_{v \text{ unassigned}} \frac{Q - 1 + q^{w_{av}}}{Q}$. All entries are rational, and the total exponent of q in any product is at most M. Products over polynomially many vertices therefore have polynomial bit length. There are Q candidate labels and Q^2 pair terms at every step. Straightforward exact numerator/denominator arithmetic already gives polynomial bit complexity; no numerical exponential or logarithm is computed by the labeling algorithm. □

Corollary 4.2 (Nearly lossless uniform padding)**.** *Let $h = \max\{2, \lceil \log_2 s \rceil\}$ and $Q = s^{O(1)}$. Suppose, uniformly over the active calls under consideration,*

$$\frac{QR}{M} = o(1), \qquad \frac{Q^2 C (\log s)^2}{M} = o(1), \qquad \frac{LQ^2}{M} = o(1). \tag{8}$$

Round the largest pair load upward to a multiple H_{pad} of L. Then

$$P := H_{\text{pad}} Q^2 = (1 + o(1))M. \tag{9}$$

Proof. Multiply (6) by Q^2 and divide by M. Each excess term tends to zero by (8); rounding adds less than LQ^2. Conversely $P \ge M$, because it contains all real edges. This proves the equality with a uniform relative error. □

4.1 Why the power-scale conditions are useful and close to sharp

With $R = C = \Delta$, the earlier second-moment proof required $Q^4\Delta = o(M)$ for nearly lossless padding. The new lemma needs only $Q^2\Delta(\log s)^2 = o(M)$, in addition to harmless rounding and the weaker row term. Thus its admissible leading packet exponent is twice as large in the symmetric setting.

There is also a scoped obstruction. A row vertex of degree R_0 forces some pair load to be at least R_0/Q, regardless of the labels, so uniform padding uses at least QR_0 slots. A parallel bundle of μ edges forces at least $Q^2\mu$ padded slots. Consequently $QR_0 = O(M)$ and $Q^2\mu = O(M)$ are necessary in worst-case examples for this particular common-pair-load architecture. Taking $\mu = C$ when $R \ge C$ witnesses the column-degree scale. Our sufficient conditions match these powers with logarithmic overhead and strict asymptotic slack. The little-oh conditions themselves are not claimed necessary, and this is not a lower bound for tensor contraction or SAT.

5 Packet counts, crossing families, and the recursive asymmetry

The exact orientation construction can be retained after replacing its label assignment. The common padded mass $P = H_{\mathrm{pad}}Q^2$ gives P/L packets. In each coordinate and on each side, a fixed symbol has exactly P/r padded ports. Therefore its real-port count is at most P/r, and a child has at most

$$2P/r = (2/r + o(1))M \tag{10}$$

wires.

Only packets with two different coordinate symbols create crossing modes. Exactly a fraction $1 - 1/r$ of the padded ordered label pairs have this property. Each packet is homogeneous for it, so at valuation level i the number of tile families is at most

$$c_i \le (1 - 1/r)\frac{P}{L}\,2^{i+1} = (2(1 - 1/r) + o(1))\,\frac{M}{D_i}. \tag{11}$$

This retains the exact equal-symbol saving as well as the improvement from padded mass $3M$ to $(1 + o(1))M$.

Lemma 5.1 (Side-degree invariant)**.** *At depth $d \ge 1$, orient the child bipartition with all retained tensors on the row side and all fresh packet tensors on the column side. If all tensors created so far have degree at most Δ, then*

$$R_d \le \Delta, \qquad C_d \le 2L_{d-1}.$$

These bounds hold regardless of the chosen mode tuple, interpolation node, or interpolation value.

Proof. The source child construction places all retained parent tensors from both old sides onto one new side. None gains a port. Each vertex on the other side is newly created for a parent packet and has at most its L_{d-1} row and L_{d-1} column ports. The degree is independent of the coefficient values and of the mode indices. The same construction repeats at every subsequent call. □

6 Precision and entropy with an additive buffer

The previous packets already sharpened the Hermite precision from a leading $24MD/u$ to $6MD/u$ [6, 7]. We use that result, with its proof ingredients explicit here, and change how successive packet exponents accommodate inherited precision.

At valuation $i < u \le a$, let $q_i = \rho(u - i)$, with $q_i = 0$ for $i \ge u$. There are $L/2^i$ nonzero grid points of valuation i. Thus the exact condition count is

$$\sum_{t \ne 0} q_t = \rho L(2u - 2 + 2^{1-u}). \tag{12}$$

Choose

$$\rho = \left\lceil \frac{M(2L - 1) + 1}{L(2u - 2 + 2^{1-u})} \right\rceil. \tag{13}$$

The denominator is an integer because $u \le a$. The count exceeds the degree of F, and when $u \to \infty$, $u/M \to 0$, $\rho = (1 + o(1))M/u$ uniformly.

The interpolation multiplier at a point of valuation i has negative valuation

$$\Lambda_i = \rho \sum_{j=1}^{a-i} j \min\{j, u - i\}\, 2^{a-i-j} \le 6\rho D_i,$$

using $\sum_{j\ge1} j^2/2^j = 6$. The inverse local Hermite factor is integral, and evaluating a jet of degree below q_i loses at most $(q_i-1)(a+1-i)$ more bits. Hence the simple sufficient precision

$$p_t = p + 6\rho D_i + \rho u(a+1) \tag{14}$$

preserves the required residue modulo 2^p. The exact smaller loss can also be used; the theorem needs only this upper envelope.

Let $\lambda = \lceil \log_2 s \rceil$ and

$$b = \lceil 6 \log_2 \lambda \rceil, \qquad a_d = a_{d-1} + u_d + b, \qquad a_{-1} = 0. \tag{15}$$

Assume active sizes are $M > s/\lambda$, the depth is $O(\log \lambda)$, and $a_d, u_d = O(\lambda)$ with $u_d \to \infty$ uniformly. For large s, $\rho \le 2M/u$. Every precision increment is therefore bounded by $O(s2^{a_j} + s\lambda)$ at its ancestor depth j. Summing gives the explicit, sufficient estimate

$$p = O(s\lambda^2 2^{a_{d-1}}). \tag{16}$$

At the root it follows from $P_0 = O(s)$. Since $i < u_d$,

$$D_i \ge 2^{a_d - u_d + 1} = 2^{a_{d-1}+b+1}.$$

Combining with $s/M \le \lambda$, $u_d \le \lambda$ gives

$$\frac{p}{MD_i/u_d} = O(\lambda^4 2^{-b}) = O(\lambda^{-2}) = o(1).$$

Also $\rho u(a+1)/(MD_i/u) = O(u(a+1)/D_i) = o(1)$. Consequently

$$p_t \le (6 + o(1)) \frac{MD_i}{u_d} \tag{17}$$

uniformly. The old cumulative rule doubled every retained range; only an additive $O(\log\log s)$ buffer is required.

6.1 The factorial weighted-partition bound

For the weights in (4),

$$\sum_{j\ge0} -\ln(1 - e^{-\theta w_j}) = \frac{\pi^2 + o(1)}{18\theta}, \qquad \theta \downarrow 0.$$

Indeed $w_j = 3j + O(\log(j+2))$; compare it outside a fixed initial segment with $(3 \pm \varepsilon)j$, use $\int_0^\infty -\ln(1-e^{-x})\,dx = \pi^2/6$, and then send ε to zero. The finite initial segment contributes $O(\log(1/\theta))$. For at most c_i families, the number of tuples of weighted degree less than p_t is bounded by

$$N_i \le \exp\left(\theta p_t + \frac{(\pi^2 + o(1))c_i}{18\theta}\right).$$

Extending a finite family to all $j \ge 0$ only increases this bound. Optimize θ using the upper envelopes for p_t and c_i in (11) and (17). The chosen θ tends uniformly to zero, since it is proportional to $\sqrt{u_d}/D_i$ and $D_i \ge \lambda^6$. We obtain

$$\ln N_i \le (\kappa_r + o(1)) \frac{M}{\sqrt{u_d}}, \qquad \kappa_r = \pi\sqrt{\frac{8}{3}(1 - 1/r)}. \tag{18}$$

No lower bound on the actual family count is needed: its prescribed upper envelope can be used directly in the exponential estimate.

7 The explicit asymmetric schedule and its global bound

For Theorem 1.1, set

$$r = 16, \qquad H = \lceil s/\lambda \rceil, \qquad b = \lceil 6 \log_2 \lambda \rceil,$$
$$u_0 = \lfloor \lambda/81 \rfloor, \qquad u_d = \left\lfloor \frac{\lambda}{108\, 4^{d-1}} \right\rfloor \ (d \geq 1), \qquad a_d = \sum_{j=0}^{d} (u_j + b). \tag{19}$$

The sum of the leading retained ranges after the root equals $\lambda/81$, so the cumulative budget is $2\lambda/81$.

7.1 Applicability without a circular depth assumption

First consider every hypothetical depth $d \leq d_* := \lceil \log_6 \lambda \rceil + 2$ and every active $M > H$. Uniformly in that range,

$$a_0 \leq \lambda/81 + O(\log \lambda), \quad a_d \leq 2\lambda/81 + O((\log \lambda)^2), \quad u_d \geq \lambda^{\eta + o(1)}, \quad \eta = 1 - \frac{\ln 4}{\ln 6} > 0.$$

Thus flooring is a uniform relative $o(1)$ perturbation.

At the root, both degree bounds are Δ. Since $Q = 16^{a_0}$,

$$Q^2 \Delta = s^{809/810 + o(1)}, \qquad \frac{8}{81} + \frac{9}{10} = \frac{809}{810} < 1.$$

The other terms in (8) are smaller. Below the root, Lemma 5.1 gives $R_d \leq \Delta$ and $C_d \leq 2^{a_{d-1}+1}$. Their relevant leading exponents are

Quantity	Upper power of s, up to $o(1)$	Exact margin
$Q_d R_d$	$4(2/81) + 9/10 = 809/810$	$1/810$
$Q_d^2 C_d$	$8(2/81) + 2/81 = 2/9$	$7/9$
$L_d Q_d^2$	$9(2/81) = 2/9$	$7/9$

These fixed gaps absorb the logarithmic factors, the additive buffers, and the factor $s/M \leq \lambda$. New packet degrees are below Δ because $2/81 < 9/10$. Corollary 4.2 therefore holds uniformly for every hypothetical active depth under consideration.

For sufficiently large s, each child has at most $M/6$ wires by (10). An active path cannot then exceed $\lceil \log_6 \lambda \rceil + O(1)$. This validates the hypothetical range and closes the induction. More sharply, for a uniform $\delta_s \to 0$,

$$M_d \leq s \left(\frac{1 + \delta_s}{8} \right)^d. \tag{20}$$

All exceptional bounded sizes, or sizes at which the explicit parameter checks fail, can be handled by direct summation. The theorem is asymptotic; the strict but small root gap does not suggest practical input thresholds.

7.2 Summing the actual work

For $r = 16$, (18) gives $\kappa_{16} = \pi\sqrt{5/2}$. The root contributes $(9 + o(1))s/\sqrt{\lambda}$ to $\sum_d M_d/\sqrt{u_d}$. The tail contributes at most

$$(1 + o(1)) \frac{s\sqrt{108}}{\sqrt{\lambda}} \sum_{d \geq 1} \frac{2^{d-1}}{8^d} = \left(\sqrt{3} + o(1) \right) \frac{s}{\sqrt{\lambda}}.$$

The perturbation in (20) changes a strictly convergent geometric series by $o(1)$. Thus

$$\sum_d \frac{M_d}{\sqrt{u_d}} \le (9+\sqrt{3}+o(1))\frac{s}{\sqrt{\lambda}}. \tag{21}$$

The polynomial number of interpolation nodes, values, and group choices at each of $O(\log\lambda)$ depths adds only $O(\lambda\log\lambda)$ to the logarithm of the total work. Direct leaves cost $2^{O(s/\lambda)}$; original entry routines cost $\exp(O(s^{9/10}))$; packet entry routines cost $\exp(O(L\log s)) = \exp(s^{2/81+o(1)})$. These are all $2^{o(s/\sqrt{\lambda})}$. Multiplying the per-call costs over the branching tree and dividing the natural logarithm by $\ln 2$ proves (1).

The total label sets and interpolation degrees are polynomial in s; in particular $M(2L-1) < s^4$ eventually. Precisions remain polynomial by (16) and $a_d = O(\lambda)$. Old descriptions are immutable and shared by reference. Enumerate tuples, interpolation values, and child groups depth first, accumulating one sum and product at a time. This preserves polynomial working space and finishes the proof of Theorem 1.1.

8 An additional improvement from the actual Cauchy row domain

The factorial Cauchy packet in the reviewed proposals uses the enlarged row domain $z \in 2\mathbb{Z}_{(2)}$ (up to a highly divisible extension-ring perturbation). The actual reciprocal rows exclude a residue class. Keeping this information gives a termwise stronger integral mode basis without adding any source-tile families. The result is local: its worst limiting slope over all valuation levels remains three.

Write $\nu_2(0) = +\infty$, and let

$$S_i = \{w \in \mathbb{Z}_{(2)} : 0 \le \nu_2(w) \le i\}, \qquad i \ge 0.$$

Define $G(b) = b + \nu_2(b!)$ for $b \ge 0$. The next construction is explicit; no theorem about generalized factorials is needed.

Lemma 8.1 (An explicit ordering of a punctured two-adic ball)**.** *For each $i \ge 0$ there are distinct positive integers $a_0^{(i)}, a_1^{(i)}, \dots$ in S_i and nondecreasing integers $F_i(0), F_i(1), \dots$ such that*

$$F_i(b) = \min_{w\in S_i} \sum_{h<b} \nu_2(w - a_h^{(i)}), \qquad \sum_{h<b} \nu_2(a_b^{(i)} - a_h^{(i)}) = F_i(b).$$

They can be constructed as follows. For $i = 0$ take $a_b^{(0)} = 2b+1$ and $F_0(b) = G(b)$. For $i \ge 1$, the sequence F_i is the sorted merge, with multiplicity, of the two strictly increasing sequences

$$G(n), \qquad n + F_{i-1}(n), \qquad n = 0, 1, 2, \dots.$$

When the next element comes from the first sequence output the point $2n+1$; when it comes from the second output $2a_n^{(i-1)}$. Resolve a tie in favor of the first sequence. For every i, b,

$$a_b^{(i)} \le 2b+1, \qquad F_i(b) \ge \nu_2(b!).$$

Proof. For the ordinary consecutive ordering $0, 1, \dots$, the minimum valuation of a product of n consecutive differences over $\mathbb{Z}_{(2)}$ is $\nu_2(n!)$. This follows by counting, for every $r \ge 1$, the at least $\lfloor n/2^r \rfloor$ factors in the appropriate residue class; equality holds at the next consecutive integer. Scaling and translating gives the assertion for odd integers, including rational two-integral arguments.

Split S_i into its odd part $1 + 2\mathbb{Z}_{(2)}$ and its even part $2S_{i-1}$. Differences between the two parts are units. If n_o odd and n_e even points have already been chosen, the least possible valuation for the next odd point is $G(n_o)$, and that for the next even point is $n_e + F_{i-1}(n_e)$. Choosing the smaller proves the asserted minimum and the merge recursion. Both component sequences are strictly increasing: G increases by at least one, as does $n + F_{i-1}(n)$.

For the lower bound, use induction on i. The second component is termwise at least G once $F_{i-1}(n) \geq \nu_2(n!)$. The sorted merge of two copies of G is precisely $\nu_2(b!)$, because $\nu_2((2n)!) = \nu_2((2n+1)!) = G(n)$. Thus the merge in the statement is no smaller. The base case is immediate.

The same termwise comparison and the odd-first tie rule imply that before an even point is selected, $n_o \geq n_e + 1$. For an odd point, $2n_o + 1 \leq 2b + 1$. For an even point, induction bounds it by $2(2n_e + 1) \leq 2(n_o + n_e) = 2b$. This proves the point-size bound and also makes the construction polynomial in the number of requested points and i. □

Lemma 8.2 (Uniform valuation slope)**.** *Put*

$$R_i = 1 + \frac{3}{4^{i+1} - 1}.$$

Then, for all $i, b \geq 0$,

$$R_i b - 4\log_2(b+2) \leq F_i(b) \leq R_i b.$$

Proof. The base case follows from $F_0(b) = 2b - s_2(b)$, where $s_2(b)$ is the binary digit sum. For the induction step put $B = 1 + R_{i-1}$, so $2 < B \leq 3$ and

$$R_i = \left(\frac{1}{2} + \frac{1}{B}\right)^{-1} = \frac{2B}{B+2}.$$

Substitution gives the displayed closed form for R_i. The two component sequences are bounded above by $2n$ and Bn. Their merge at index b is at most $R_i b$: the number of terms in these two arithmetic sequences no greater than $R_i b$ is at least $b + 1$, since the two unrounded ratios sum to the integer b.

For the lower bound let $x = F_i(b)$ and $L_b = \log_2(b+2)$. Each of the first $b+1$ terms of the merge has a component index at most b. The lower bounds for those components are $2n - L_b$ and $Bn - 4L_b$. Consequently

$$b + 1 \leq \frac{x + L_b}{2} + 1 + \frac{x + 4L_b}{B} + 1,$$

so

$$x \geq R_i b - R_i - \left(\frac{R_i}{2} + \frac{4R_i}{B}\right) L_b.$$

Here $R_i \leq 6/5$ and $R_i/2 + 4R_i/B = (B+8)/(B+2) \leq 5/2$. Since $L_b \geq 1$, the last two losses together are at most $4L_b$. This proves the uniform estimate. □

Lemma 8.3 (Stability under an unramified extension perturbation)**.** *Let $\mathcal{O}$ be the unramified coefficient ring of the contraction argument, with valuation extending ν_2. Suppose $R \geq 2$, $w_0 \in S_i$, $w - w_0 \in 2^R\mathcal{O}$, and $0 \leq b < 2^{R-2}$. Then*

$$\nu_2\left(\prod_{h<b}(w - a_h^{(i)})\right) \geq F_i(b).$$

Proof. The assertion is immediate for $b = 0$. Otherwise the selected integers are distinct and satisfy $1 \le a_h^{(i)} \le 2b - 1 < 2^{R-1}$. Choose the ordinary residue $c \in \{0, \dots, 2^R - 1\}$ of w_0. If c is nonzero and is not a selected point, take $w' = c$. It belongs to S_i, and all factor valuations are below R and unchanged by the perturbation. If $c = 0$, then $\nu_2(w_0) \ge R$, so $i \ge R$; take $w' = 2^R \in S_i$. Again all differences to the selected points have valuation below R. Finally, if $c = a_h^{(i)}$ for one selected point, take $w' = a_h^{(i)} + 2^R \in S_i$. Its difference from that point has valuation R, and its other difference valuations agree with those of w; the corresponding difference for w has valuation at least R. In each case the product valuation for w is at least that for $w' \in S_i$, which is at least $F_i(b)$ by Lemma 8.1. This argument uses only ordinary residues of w_0; w itself may have nonzero extension coordinates. □

Theorem 8.4 (Annulus Cauchy modes)**.** *At a contraction node of valuation i, retain the existing packet layout and column tiles. Put*

$$\alpha = 2^{a+1}, \quad T = t + \alpha\zeta, \quad R = a + 1 - i, \quad J = 2^{a-i-1}, \quad y = y_0 + 2^{i+1}j,$$

where $0 \le j < J$ and $\zeta \in \mathcal{O}$. For a row crossing that tile, let

$$d_x = T - y_0 + x, \quad z_x = \frac{2^{i+1}}{d_x}, \quad A_x = \frac{\mathcal{C}(T)}{\mathcal{C}'(0)d_x}, \qquad \xi_b = 2a_b^{(i)}.$$

Define

$$r_b(z) = \frac{b! \prod_{h<b}(z - \xi_h)}{\prod_{h=0}^{b}(1 - hz)}, \qquad s_b(j) = \frac{(1 - b\xi_b)\binom{j}{b}}{\prod_{h=0}^{b}(1 - \xi_h j)}.$$

The exact finite identity

$$W_{xy}(T) = A_x \sum_{b=0}^{J-1} r_b(z_x) s_b(j)$$

holds. All source functions $s_b(j)$ are two-integral; all displayed new denominators are units. The target coefficient $A_x r_b(z_x)$ is divisible by $2^{\omega_i(b)}$, where

$$\boxed{\omega_i(b) = R + b + F_i(b) + \nu_2(b!)}.$$

In particular these weights are termwise no smaller than the previous factorial weights with the same base contribution R, and

$$\omega_i(b) = R + \left(3 + \frac{3}{4^{i+1} - 1}\right) b + O(\log(b + 2)),$$

with an absolute error constant. For $i = 0$ the exact formula is $\omega_0(b) = R + 2b + 2\nu_2(b!)$.

Proof. The crossing condition gives $e = \nu_2(t - y_0 + x) \le i$. Since $\alpha\zeta$ has valuation at least $a + 1$, one has $\nu_2(d_x) = e$, so z_x is divisible by two and $\nu_2(A_x) \ge a + 1 - e \ge R$, as in the original integrality proof. For

$$w = z_x/2, \qquad w_0 = 2^i/(t - y_0 + x),$$

one has $w_0 \in S_i$ and

$$\nu_2(w - w_0) \ge i + a + 1 - 2e \ge a + 1 - i = R.$$

Lemma 8.3 therefore yields

$$\nu_2\left(\prod_{h<b}(z_x - \xi_h)\right) \ge b + F_i(b) \qquad (b < J = 2^{R-2}).$$

The remaining denominator factors in r_b are units, proving the weight bound after multiplying by $b!$ and A_x.

For completeness the identity is a telescoping Cauchy elimination. Let

$$U_b(z) = \prod_{h<b} \frac{z-\xi_h}{1-hz}, \qquad V_b(j) = \prod_{h<b} \frac{j-h}{1-\xi_h j}.$$

Then direct multiplication gives

$$\frac{U_b(z)V_b(j) - U_{b+1}(z)V_{b+1}(j)}{1-zj} = \frac{(1-b\xi_b)U_b(z)V_b(j)}{(1-bz)(1-\xi_b j)} = r_b(z)s_b(j).$$

For $0 \le j < J$ the factor $V_J(j)$ vanishes. Summation gives $1/(1-zj) = \sum_{b<J} r_b(z)s_b(j)$, which becomes the kernel identity after multiplication by A_x. Since every ξ_b is even, all source denominators and $1-b\xi_b$ are units. The binomial coefficient is evaluated as an ordinary integer: no factorial is divided in the residue ring. The comparison with the earlier weights uses $F_i(b) \ge \nu_2(b!)$, and the slope follows from Lemma 8.2 and $\nu_2(b!) = b - s_2(b)$. □

Algorithmic consequence and its scope. Use one mode for each existing packet, tile, and index $b < J$, with source linear form $\sum_j s_b(j) X_{y_0+2^{i+1}j}$. Its source and target groups are unchanged. Hence the number of families, child network sizes, entry-evaluation mechanism, and canonical-lift invariant are unchanged. Truncate by the new exact weights. The admissible tuple set is contained in that certified by the old factorial weights, for every finite parameter choice. Precomputing the ordering and evaluating the integral features costs only a polynomial factor in the existing parameters. The uniform entropy analysis may still use slope three, while a fixed valuation level has the sharper slope $\gamma_i = 3 + 3/(4^{i+1}-1)$. These slopes approach three as i grows. Thus this result by itself is not a new logarithmic power in the worst-case SAT exponent, nor does it prove a uniform leading constant strictly better than the slope-three envelope over all levels. A valuation-dependent taper may exploit the local improvements, but that optimization is not assumed in the global theorem here.

9 Consequences for SAT and integration with the earlier results

For bounded clause density, the source uses a linear-size bounded-degree network. A clause contributes its satisfaction tensor. A variable's occurrences are linked by a tree of equality tensors, and an additional subdivision makes the network bipartite. Every satisfying assignment has one consistent extension to the internal equality wires. Unused variables contribute a power of two. Contracting modulo 2^{n+1} recovers the exact integer count because it lies in $[0, 2^n]$.

The new packing and schedule preserve this encoding and exactness. Moreover the low-degree theorem in Appendix A applies to the original bounded-degree networks. The guaranteed time is still $\mathrm{poly}_k(n,m)2^{O_{k,D}(n/\sqrt{\log n})}$ when $m \le Dn$. The numerical coefficient in terms of variables additionally depends on the wire count of the chosen encoding; the wire-level constants in this report must not be substituted directly for it.

For arbitrary density, the earlier reductions use sparse parity tests and an incidence subset representation with $O_k(n \log\log n)$ wires. The uniform density-aware packet improves that count in its stated regimes, yielding (2). The scalar parity packet reduces the retained incidence-degree cap and removes the auxiliary coefficient axis for that lift. These changes are compatible with our engine: it permits $\ell = 1$ as well as truncated extension rings, and its inputs are still arbitrary tensors

meeting the stated oracle bounds. Their correctness and density constructions remain separate imported results. This report adds no new sparsification or expander-code theorem.

For integration, replace the preprint's collision labeling by Theorem 4.1, retaining the same packet orientation. Keep the exact side provenance of every child; do not discard it into a single maximum-degree parameter. Use (19), the taper (13), and the precision rule (14). Replace the mode basis by either (4) or the annulus basis. The earlier streamed coefficient-extraction routine is also compatible, provided its temporary extra precision does not change the tensor's creation-time representatives. A separate `INTEGRATION.txt` gives the source-label map.

10 Executable evidence and its limits

The deliverable contains original standard-library Python components for pair-label balancing, packet reconstruction, annulus orderings and exact Cauchy identities, and schedule arithmetic. The recorded run summary is included as `results/verification_summary.json`. The tests use exact integers, rational numbers, and modular arithmetic for mathematical identities. Decimal constants are displays of exact formulas, not inputs to a proof or to the labeling algorithm.

The packing prototype computes every candidate conditional expectation as a rational, checks the averaging identity, chooses a minimum, and checks the final potential. Multiedges and asymmetric degree fixtures are included. An independent reconstruction counts edges by their returned labels; a packet checker verifies real-edge preservation, shared slots, coordinate orientations, group load bounds, and the unequal-packet fraction on small examples.

The annulus checks exercise selected-point minimality, the merge recurrence, the point-size and slope bounds, rational Cauchy identities, and arithmetic over

$$(\mathbb{Z}/2^{160}\mathbb{Z})[U]/(U^2+U+1).$$

The nontrivial extension coordinates are essential: arbitrary extension residues do not satisfy ordinary integer-valued binomial divisibility. The proof uses the actual row's highly divisible congruence to a rational point in the permitted annulus.

For a concrete comparison of the new modes, use one tile family with $J = 16$, base valuation $R = 6$, and precision $p = 64$. Compare with the previous factorial weights using the *same* base contribution R. Exact dynamic programming gives:

Valuation level i	Previous feasible tuples	Annulus feasible tuples
0	1726	634
1	1726	1421
2	1726	1683
3	1726	1723

These are counts in a specified abstract mode-budget model, not running times or numbers of necessarily nonzero SAT contributions. The choice $a = i + 5$ makes the displayed J and R consistent with the packet geometry.

The schedule checker verifies the exact rational exponent margins, budget identities, convergence conditions, and numerical evaluations of the displayed constants. Its finite parameter examples are labeled as diagnostics; many small logarithmic sizes fail the asymptotic conditions and are not represented as successful practical instances.

These are component checks and proof-supporting fixtures. The all-size conclusions come from the proofs. There is no full SAT engine in the archive, no SAT performance benchmark, no compiled Lean or Rocq theorem, and no verification of the entire upstream dependency graph.

11 Why the logarithmic order survives, and further research

The weighted enumeration still has linearly growing mode weights. With $c \asymp M/D$, $p \asymp MD/u$, and $w_j \asymp j$, choose B to be a sufficiently small fixed multiple of $D/\sqrt{u}$. There are $\Theta(cB) = \Theta(M/\sqrt{u})$ mode variables in indices $[B, 2B)$. Every zero-one choice among them fits the budget after adjusting the fixed multiple. Thus the abstract mode pool can have $\exp(\Omega(M/\sqrt{u}))$ feasible tuples. Their total unweighted degree is $O(M/\sqrt{u}) \le M$, so imposing the affine-product degree cap alone does not remove the obstruction. This is a scoped enumeration example, not a statement that every such tuple is nonzero on every actual SAT network, and not a lower bound for all algorithms.

The annulus slopes tend to three as i grows. Consequently a uniform replacement of three by a larger fixed slope has not been obtained. The new packing conditions still constrain u to $O(\log s)$ in this architecture. Together these observations explain why the present result changes the leading coefficient while preserving the square-root logarithmic denominator.

The most concrete next questions are:

1. **Can feasible modes be summed without enumerating them?** A useful theorem must provide an actual uniform factorization or coefficient algorithm for the packet-group tensors, with bounded wire count and controlled oracle complexity. Merely assigning shorter names to the tuples is insufficient.

2. **Can the dense incidence representation lose its extra logarithm?** An $O(n)$-wire representation, with valid degree and entry-cost bounds, would remove the current $\log\log n$ overhead. The earlier packets state this missing interface; the present work does not construct it.

3. **Can instance-sensitive degree and multiplicity information improve the first call?** The root alone pays the symmetric high-degree constraint. A construction that uses smaller parallel multiplicity, local neighborhood balance, or large-degree special structure could improve this remaining coefficient bottleneck. It would need a polynomial-time construction, not just a random-existence estimate.

4. **What is the best finite taper with annulus weights?** The exact sequences $F_i(j)$ give more information than their limiting slopes. A finite optimization that incorporates actual family counts, node losses, and inherited precision could reduce enumeration in the specified finite model. It must not be advertised as a new logarithmic order unless a uniform asymptotic argument establishes that.

5. **Can the new pieces be formalized and integrated?** The rational conditional-expectation lemma, packet invariants, and finite Cauchy identity are distinct formalization targets. A full result also needs the cost model, exact quotient-ring arithmetic, inherited precision, and immutable tensor descriptions. The supplied prototypes are a starting specification, not substitutes for those proofs.

A A low-degree contraction coefficient

When the root has maximum degree at most $\lceil s^{1/12} \rceil$, one can instead maintain that same small degree bound throughout and use symmetric packing. Set

$$r = 48, \qquad u_d = \left\lfloor \frac{\lambda}{14\, 8^d} \right\rfloor, \qquad a_d = \sum_{j \le d} (u_j + b), \quad b = \lceil 6 \log_2 \lambda \rceil, \quad H = \lceil s/\lambda \rceil.$$

Then $a_d \le 4\lambda/49 + O((\log\lambda)^2)$. The new packet degree is at most $s^{4/49+o(1)} < s^{1/12}$. Since $3^5 < 2^8$, one has $\log_2 48 < 28/5$, and

$$\frac{8}{49}\log_2 48 + \frac{1}{12} < \frac{8}{49}\frac{28}{5} + \frac{1}{12} = \frac{419}{420} < 1.$$

Thus $Q^2\Delta\,\mathrm{polylog}(s) = o(M)$ and $LQ^2 = o(M)$ uniformly at active calls. Coarse shrinkage $M/12$ and hypothetical depth $\lceil \log_{12}\lambda\rceil + 2$ give the same noncircular bootstrap; u_d grows at least as a fixed positive power of λ there. The sharp child ratio is $(1+o(1))/24$, the family coefficient is $47/24$, and the natural-log mode coefficient is $\pi\sqrt{47/18}$. Therefore

$$K_{\mathrm{low}} = \frac{\pi\sqrt{47/18}\sqrt{14}}{\ln 2(1-\sqrt{8}/24)} = 31.06413206299\ldots. \tag{22}$$

Original entry routines may still have the source's allowed $\exp(O(s^{9/10}))$ cost: that remains lower order. Bounded-degree clause networks satisfy the smaller graph-degree premise eventually. The lower coefficient is not asserted for arbitrary high-degree inputs.

For a direct intermediate comparison under the generic $s^{9/10}$ degree premise, replace 14 in this schedule by 128. Then $a_d \le \lambda/112 + O((\log\lambda)^2)$ and $2\log_2(48)/112 + 9/10 < 1$. The coefficient becomes $K_{\mathrm{sym}} = 93.9291064374\ldots$, the symmetric row of the comparison table. The main theorem gains further by using the fresh side's smaller degree instead of imposing this generic bound on both sides.

B Optimization of the depth allocation

For a fixed child ratio $c \in (0,1)$ and budget $\sum_d u_d \le A\lambda$, the continuous minimization problem is

$$\min \sum_{d\ge 0} \frac{c^d}{\sqrt{u_d}}.$$

Hölder's inequality, applied to $c^{2d/3} = (c^d/\sqrt{u_d})^{2/3}u_d^{1/3}$, gives

$$\sum_d \frac{c^d}{\sqrt{u_d}} \ge \frac{1}{\sqrt{A\lambda}(1-c^{2/3})^{3/2}}.$$

Equality is attained by $u_d = A\lambda(1-c^{2/3})c^{2d/3}$. This optimizes a stated schedule model, not all contraction algorithms.

For the asymmetric high-degree model, the root budget is $A_0 = (1-\gamma)/(2\log_2 r)$ and the retained-side cumulative budget below the root is $2A_0$, provided the fresh-side and rounding constraints also hold. When $c^{2/3} < 1/2$, the root cap is active. Spend A_0 at the root and allocate the other A_0 to later levels by the same rule. The path coefficient is

$$\frac{1}{\sqrt{A_0}}\left(1 + \frac{c}{(1-c^{2/3})^{3/2}}\right).$$

For $\gamma = 9/10$ and $r = 16$, $c = 1/8$ and $c^{2/3} = 1/4$. The root/tail shape in (19) is exactly this shape, with strict inward shift from denominator 80 to 81. The boundary coefficient is $76.4327736858\ldots$; it is an infimum at an excluded boundary, not the displayed certified coefficient.

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